

5) reescreva a EDO, com séries:

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 2x \sum_{n=0}^{\infty} C_n x^n = 0 \rightarrow$$

$$\sum_{n=2}^{\infty} n(n-1)C_n \overset{(n-2)}{\downarrow} x^{\phantom{n-2}} - \sum_{n=0}^{\infty} 2C_n \overset{(n+1)}{\downarrow} x^{\phantom{n+1}} = 0 \rightarrow$$

$k = n-2 \qquad k = n+1$

reescreva em termos de  $k$ :

$$\sum_{k=0}^{\infty} (k+2)(k+1)C_{k+2} x^k - \sum_{k=1}^{\infty} 2C_{k-1} x^k = 0 \rightarrow$$

remova um termo do ( $k=0$ ):

$$2C_2 + \sum_{k=1}^{\infty} (k+2)(k+1)C_{k+2} x^k - \sum_{k=1}^{\infty} 2C_{k-1} x^k = 0$$

junte as séries:

$$C_2 + \sum_{k=1}^{\infty} ((k+2)(k+1)C_{k+2} - 2C_{k-1}) x^k = 0 \quad \therefore C_2 = 0$$

o termo de recorrência:  $C_{k+2} = \frac{2C_{k-1}}{(k+2)(k+1)}$

Calcular os termos:

$$n \quad \boxed{K=0}$$

Para  $K$ :

$$1 \quad C_3 \quad \frac{2C_0}{3 \cdot 2} = \frac{C_0}{3}$$

$$2 \quad C_4 \quad \frac{2C_1}{4 \cdot 3} = \frac{C_1}{3 \cdot 2}$$

$$3 \quad C_5 \quad \frac{2 \cdot C_2}{5 \cdot 4} = 0$$

$$4 \quad C_6 \quad \frac{2C_3}{6 \cdot 5} = \frac{\left(\frac{C_0}{3}\right) \cdot 2}{6 \cdot 5} = \frac{C_0}{3 \cdot 3 \cdot 5} = \frac{C_0}{45}$$

$$5 \quad C_7 \quad \frac{2C_4}{7 \cdot 6} = \frac{2\left(\frac{C_1}{6}\right)}{7 \cdot 6} = \frac{C_1}{3 \cdot 7 \cdot 6} = \frac{C_1}{126}$$

$$6 \quad C_8 \quad \frac{2 \cdot C_5}{8 \cdot 7} = 0$$

Logo a relação é:  $y = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + \dots$ , ou

$$y = C_0 + C_1x + 0 + \frac{C_0}{3}x^3 + \frac{C_1}{6}x^4 + 0 + \frac{C_0}{45}x^5 + \frac{C_1}{126}x^7 + \dots$$

ou

$$y = C_0 \left( 1 + \frac{1}{3}x^3 + \frac{1}{45}x^5 + \dots \right) + C_1 \left( x + \frac{1}{6}x^4 + \frac{1}{126}x^7 + \dots \right)$$